Monitoring and controlling service performance and efficiency in dynamic datacenters

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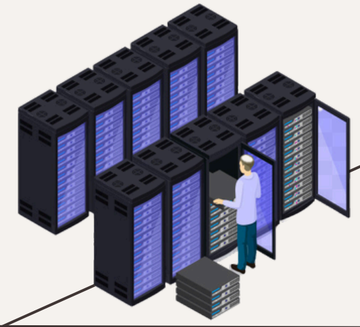

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01
Context

01

Context

- Modern datacenters are highly dynamic environments;
- Elastic and distributed services increase system complexity;
- This leads to variability, resource contention, and cascading failures
- Intelligent observability is needed for early detection and automated remediation

02
Problem

02

Problem

Delayed Detection
Complex Root Cause
PERFORMANCE ISSUES
1010101
1010101
Data Overload
No automation

03
Goals

03

Goals

Architecture

Design a multi-layer observability architecture.

Metrics

Integrate real-time metrics collection.

Testing

Implement fault injection and stress testing.

Data and ML

Use ML for anomalies, prediction, and bottlenecks.

Visualization

Provide a monitoring dashboard.

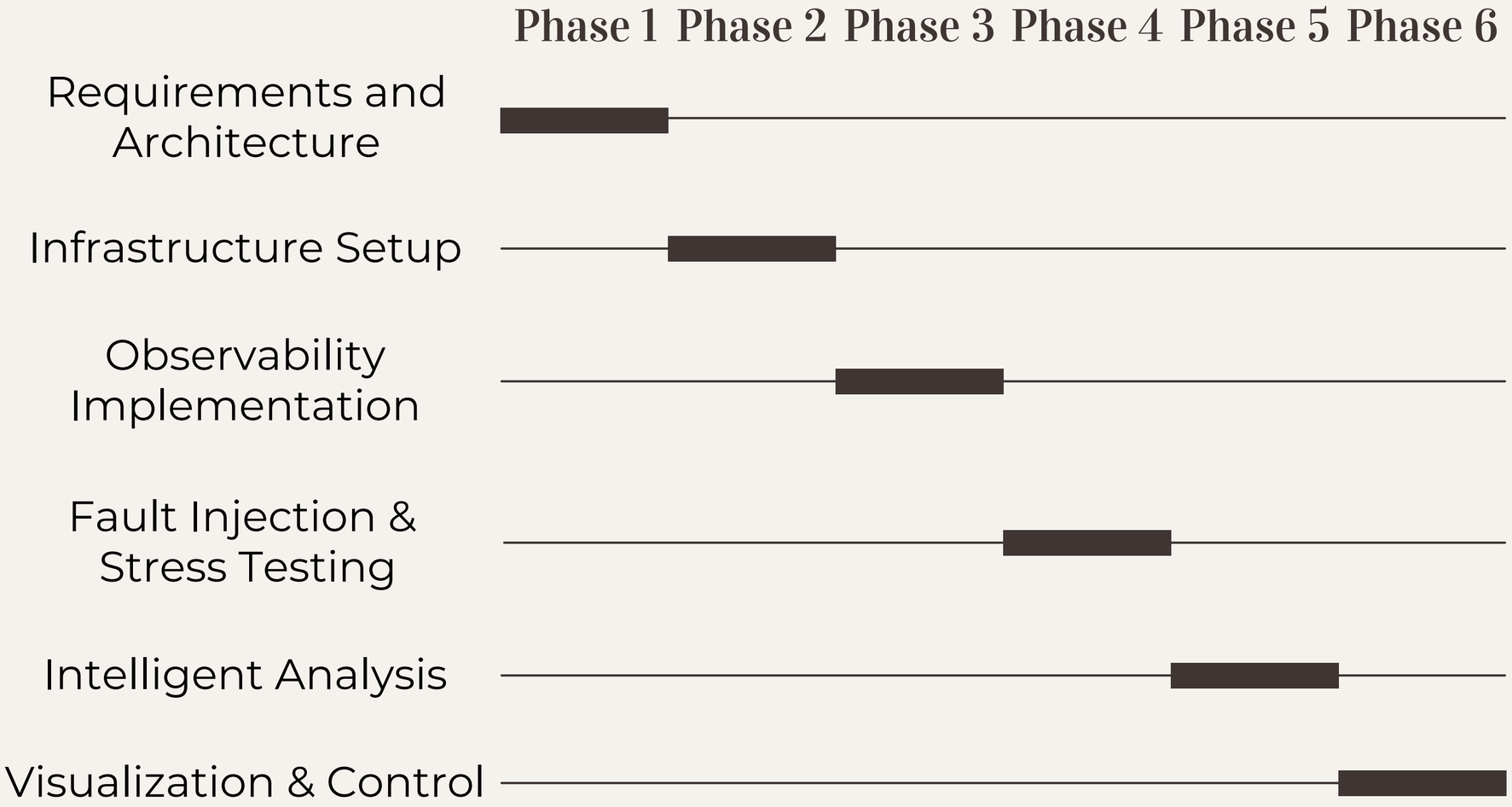
Advanced Goal (*)

Implement automated mitigation strategies.

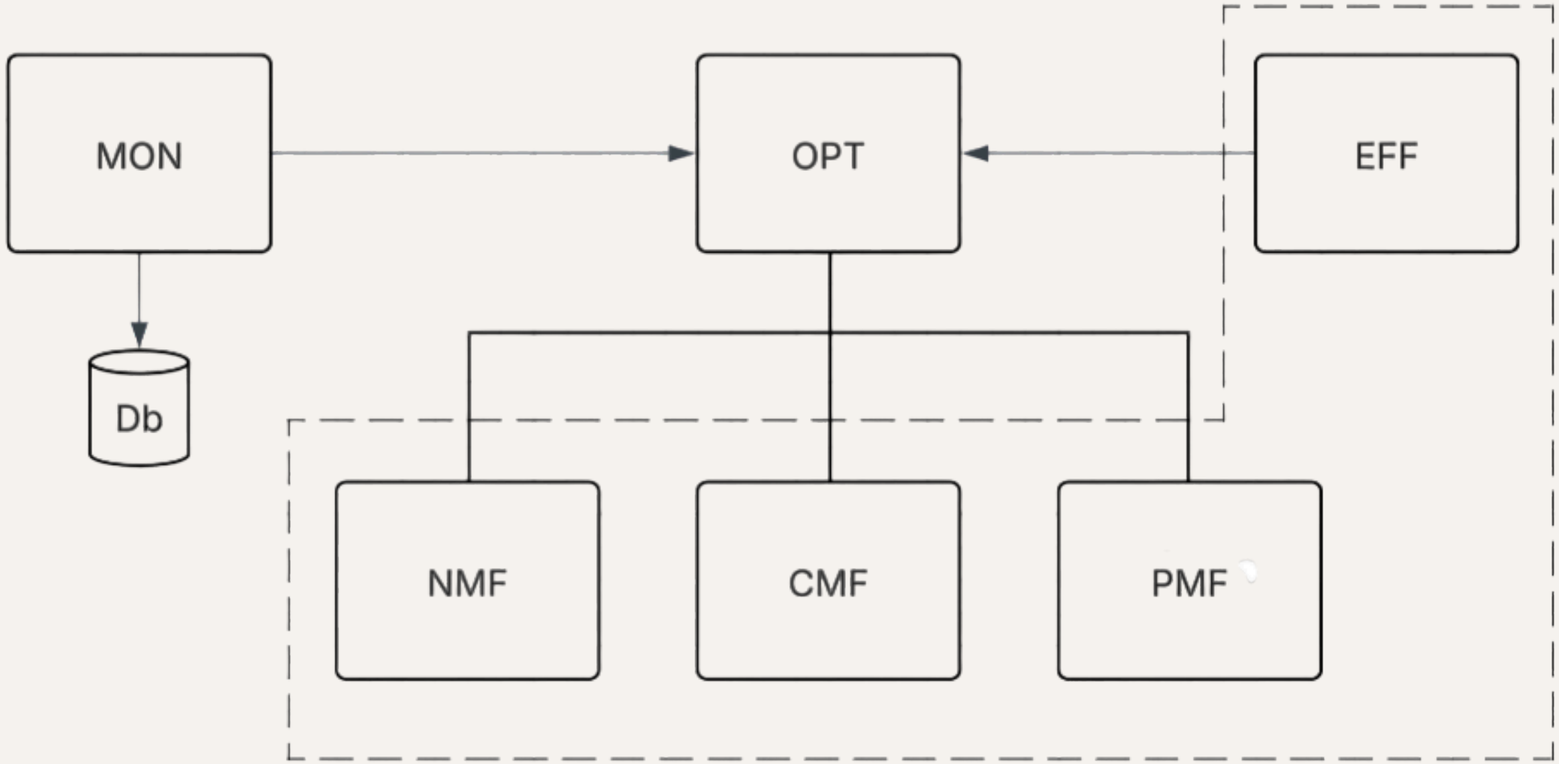
(*) = Optional

04
Tasks

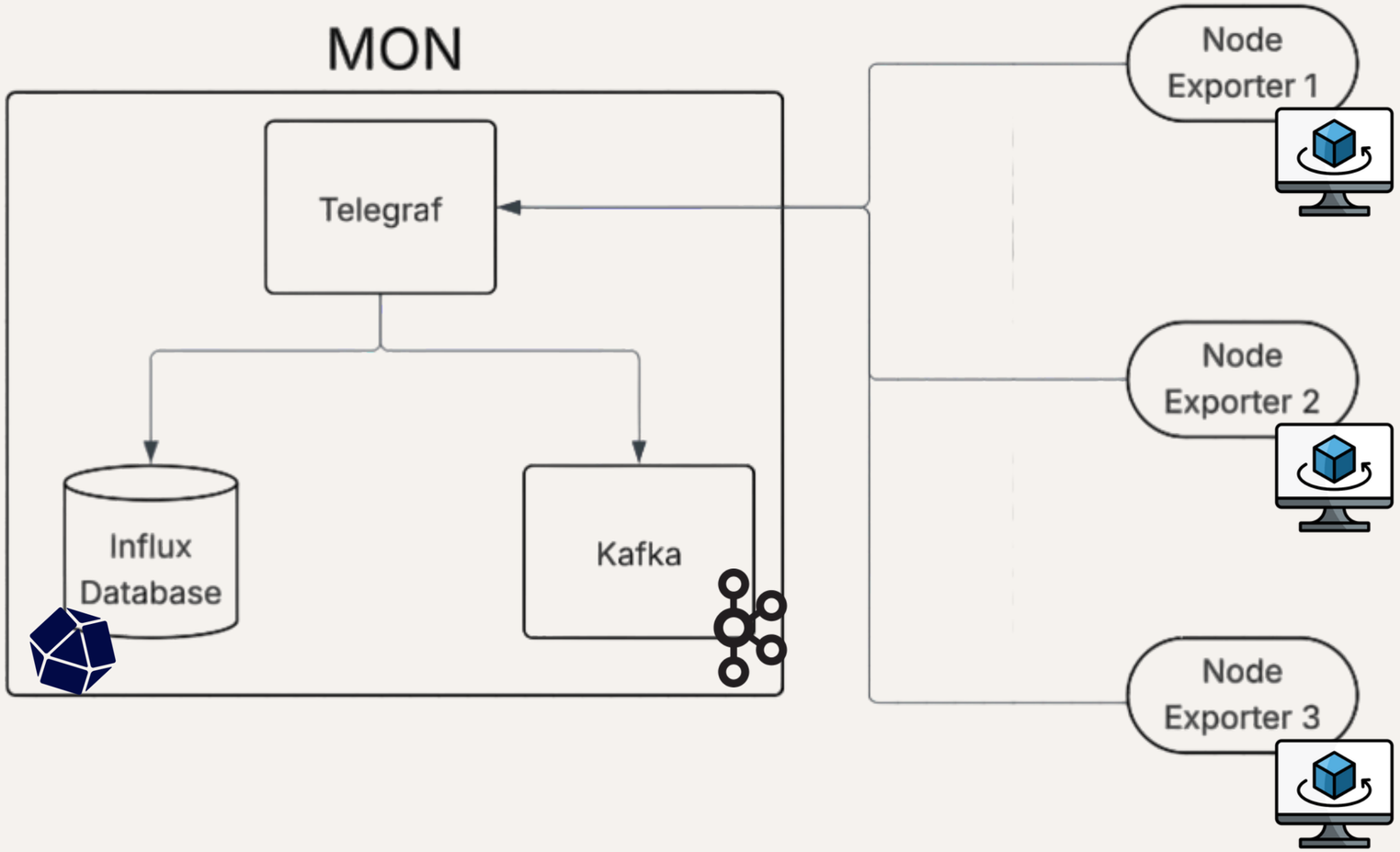
Tasks


Phase 1
Phase 2
Phase 3
Phase 4
Phase 5
Phase 6
Requirements and Architecture
Infrastructure Setup
Observability Implementation
Fault Injection & Stress Testing
Intelligent Analysis
Visualization & Control

Proposed Global System Overview



MON Module – Architecture Details



05
Expected Results

Expected Results

Experimental Validation

Validated system under controlled fault injection scenarios.

Performance Metrics

- Detection latency
- Accuracy
- Overhead introduced by monitoring

Functional Prototype

Working monitoring and control system prototype.

Documentation

Complete technical report and system documentation.

06
Technologies

Technologies

Monitoring and Observability

Virtualization and Orchestration

Fault Injection

Data Analysis and ML

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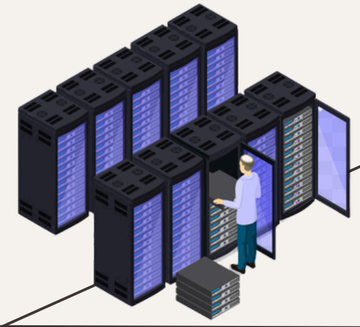
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